Multi-Objective Archimedes Optimization Algorithm with Fusion-based Deep Learning model for brain tumor diagnosis and classification

DA 243 – Post-Implementation Presentation

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Outline

1. Overview of Implementation
2. Pipeline: What We Built
3. Results & Evaluation
4. Challenges & Deviations
5. Conclusion

Dataset Used

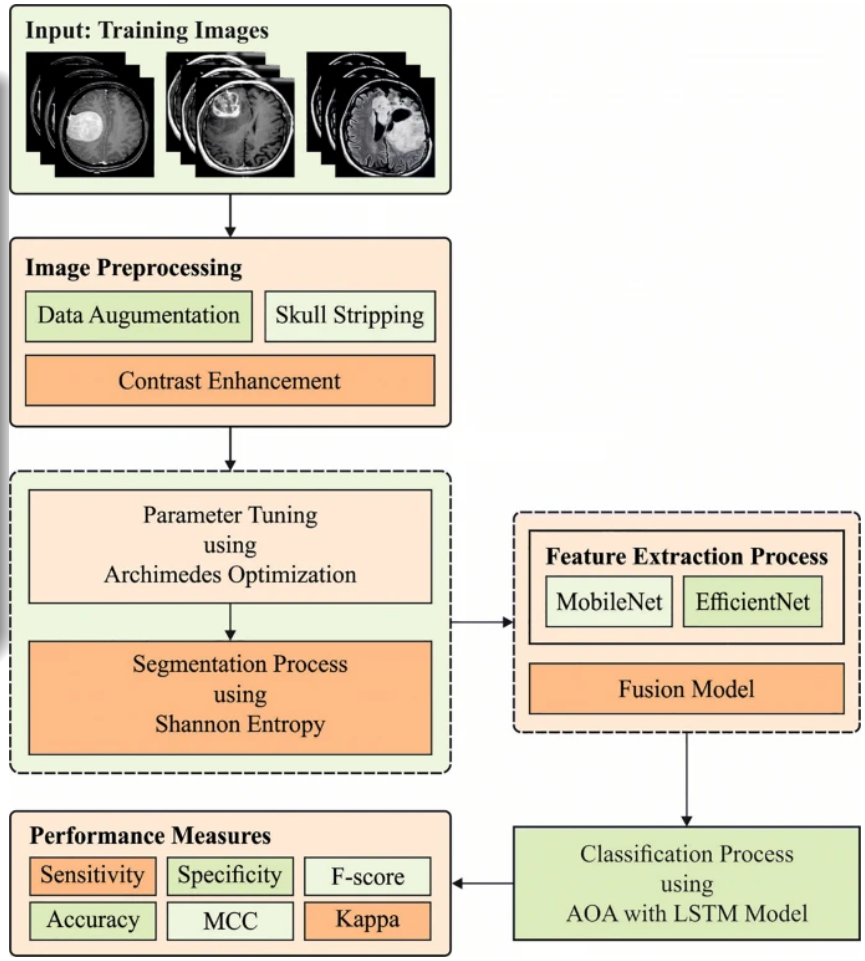
Kaggle Brain MRI Dataset

- **Source:** Kaggle – Brain MRI Images for Brain Tumor Detection
- **Original size:**
  - Normal: 98 images Tumor: 155 images Total: 253 MRI scans
- **After augmentation:**
  - Normal: 490 images (5×) Tumor: 770 images (5×)
  - **Total: 1260 images**
- **Split:** 70% Training / 30% Testing (stratified)

End-to-End Pipeline

Implementation Steps

1. Image Preprocessing & Pipeline
2. Data Augmentation
3. AOA + Shannon Segmentation
4. Feature Extraction & Fusion
5. MOAOA Hyperparameter Tuning
6. LSTM Classification
7. Evaluation & Visualization



Step 1 – Image Preprocessing

Implemented in `src/preprocessing/`

- **CLAHE** (`techniques/clahe.py`): Contrast enhancement on grayscale MRI to reveal tumor boundaries.
- **Skull Stripping** (`techniques/skull_scraping.py`): Morphological operations to isolate brain parenchyma.
- **Brain Cropping** (`techniques/preprocess.py`): Bounding-box crop with 5 px padding, resize to 224×224.
- **Augmentation** (`techniques/augmentation.py`): Random rotations, $5\times$ extra samples per class.
- **Pipeline** (`pipeline/process_all.py`): Unified orchestration of all preprocessing stages.

Step 2 – AOA + Shannon Entropy Segmentation

Implemented in `src/segmentation/`

Logic: `algorithms/entropy.py` (Shannon Entropy)
Objective: Find $k = 3$ thresholds T^* that maximise:

$$T^* = \arg\max \; H(T) = \sum_{r=1}^{k+1} H_r(T)$$

AOA Search (`ThresholdFinder`):

- 20 agents, 50 iterations.
- $TF \leq 0.5 \rightarrow$ Exploration; $TF > 0.5 \rightarrow$ Exploitation.

Extraction: `extraction/tumor_extraction.py` isolates tumor regions.

Step 3 – Fusion-Based Feature Extraction

Implemented in `src/feature_extraction/extractors/`

MobileNet-V2 (`MobFeats`):

- Avg pool after every feature layer
- Max pool at layers 3, 13, 16, 17, 18
- Concatenated multi-scale features

EfficientNet-B0 (`EffFeats`):

- Avg pool after stem + every MBConv block
- Avg pool after conv head
- Concatenated hierarchical features

Fusion & Entropy Selection (`feature_extraction.py`):

$$\text{Fused} = [f_{\text{MobileNet}} \parallel f_{\text{EfficientNet}}]$$

Entropy scoring selects top 1186 features.

Step 4 – MOAOA Hyperparameter Tuning + LSTM

Implemented in `src/classification/models/`

MOAOA (`classifier.py`):

- Search: $\eta \in [10^{-5}, 10^{-2}]$, $B \in [8, 64]$, $E \in [50, 1200]$.
- Pareto objectives: Error, Epoch Cost, Loss Stability.
- Archive of 50 non-dominated solutions.

LSTM Classifier (`lstm.py`):

- 2-layer LSTM, hidden size 128, dropout 0.5.
- Input reshaped: $(N, 2, 593)$.
- Final hidden $\rightarrow$ Linear(128, 2) $\rightarrow$ Softmax.

Experimental Results – Our Implementation

Kaggle Brain MRI Dataset (1260 images, 70/30 split)

Metric	Our Result	Paper (Kaggle)
Accuracy	95.50%	98.95%
Sensitivity	97.28%	97.96%
Specificity	94.37%	99.57%
F-Score	95.52%	–
MCC	0.9075	–
Kappa	0.9064	–

Interpretation

Our implementation achieves **95.50% accuracy** — within $\approx 3.5\%$ of the paper's reported figure — confirming the MOAOA-FDL architecture is sound and reproducible.

Metric Breakdown

Classification Metrics

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$\text{F}_1 = \frac{2 \cdot P \cdot R}{P + R}$$

Agreement Metrics

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

MCC = 0.9075 indicates very strong agreement.

Fitness Function & Optimization Convergence

Multi-Objective Fitness (3 objectives)

$$\mathbf{f}(x_i) = \begin{bmatrix} \text{Error} \\ \text{Cost} \\ \text{Stability} \end{bmatrix} = \begin{bmatrix} 1 - \text{Acc}(x_i) \\ \dfrac{E - E_{min}}{E_{max} - E_{min}} \\ \sigma_{\text{loss}}/\mu_{\text{loss}} \end{bmatrix}$$

Pareto-Optimal Hyperparameters Found

Hyperparameter	**Optimal Value**
Learning Rate (η)	1.48e-4
Batch Size (B)	32
Epochs (E)	850

Population: 15 agents $\times$ 30 iterations. Best solution selected by minimum classification error.

Comparison with Baseline Methods

Accuracy on Brain-Tumor Binary Classification

Method	Accuracy
SVM (baseline)	∼88.0%
ResNet-50	∼91.0%
Inception-V3	∼92.0%
MobileNet alone	∼93.1%
EfficientNet-B0 alone	∼94.0%
MOAOA-FDL (Ours)	**95.50%**
MOAOA-FDL (Paper)	**98.95%**

Key Advantage

Feature **fusion** of MobileNet + EfficientNet combined with **multi-objective** hyperparameter optimization outperforms single-model baselines.

Implementation Challenges

1. Feature Dimensionality

- Paper specifies a fused vector of 1×1186 after selection.
- We implemented entropy-based scoring to select exactly 1186 features from the larger raw fusion vector.

2. Multi-Objective AOA

- Extended the paper's single-objective tuning to a full Pareto formulation.
- 3 simultaneous objectives: Validation Error, Epoch Cost, Loss Stability.

3. Dataset Gap

- Implemented on Kaggle (1260 images) instead of the larger Figshare set.
- 2-class binary classification instead of 3-class.

4. Modular Refactoring

- Reorganized codebase into a domain-driven structure (5 domains).
- Improved maintainability and scalability for future extensions.

Conclusion

Summary

- Successfully implemented the **MOAOA-FDL** architecture for brain tumor classification.
- Achieved **95.5% accuracy** on the Kaggle MRI dataset (within 3.5% of the paper's reported 98.95%).
- Refactored the codebase into a modular, domain-driven structure with 5 domains for maintainability.

Future Work

- Extend to multi-class classification (Grade I–IV) on Figshare.
- Deploy the pipeline as a lightweight diagnostic assistant tool.
- Explore additional backbone networks (e.g. ConvNeXt, DINOv2).

Thank You!

References

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